Which Model Changes Can Counterfactual Explanations See? A Ground-Truth Benchmark

[author names and affiliation withheld for review]

Abstract. Counterfactual methods are used to explain how a classifier changed after an update, but without ground truth one cannot tell which method explains which kind of change. We inject known changes into classifiers on rendered images with exact factors, including a grid that varies the rotation and the offset of the decision boundary, and compare counterfactual differencing, gradient differences, importance differences, disagreement search and differential counterfactuals (Δ-CE), minimal inputs on which exactly one model changes its decision by a margin. A linear analysis predicts that gradient differences see a rotation and are blind to an offset. Both halves were confirmed on CNNs in tests registered before the grid was run (Recovery 0.08 on offsets; +0.57 from the smallest to the largest rotation), and the side of the boundary on which witnesses exist follows the sign of the offset. Margin witnesses abstain on null updates, on rendered images and on a real tabular dataset, where counterfactual differencing attributes a change to about 99% of points. After projection, witnesses and gradient differences differ when the old model's normal is taken at the query; taken at the witness, the projected Recovery gap narrows from -0.19 to $+0.03$ [0.01, 0.05] (a post-hoc explanation confirmed on new data, seen on two designs). A pre-registered hypothesis in favour of Δ-CE was rejected.

1 Introduction

Deployed classifiers are updated routinely, and an update that improves accuracy can still change individual decisions in unwanted ways [11]. Explaining *how* the decisions of f_{new} differ from those of f_{old} is therefore a natural goal of explainable AI under distributional change [3, 12]. Artelt et al. [1, 2] compare a counterfactual [8] of the same input under each model, or approximate this by a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which two models disagree. These families assume different things about *what* changed (a direction of the weights, a boundary, a region of disagreement), and without ground truth these assumptions go unchecked.

We ask which kind of change each family can see, using injected changes with exact ground truth. We also include *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search [5, 6] at a query; unlike robust counterfactuals [9, 10], which seek a point valid for both models, it seeks one valid for exactly one, and the margin makes it return nothing when two models differ only by jitter.

Contributions. (i) A benchmark with exact ground truth for explaining model updates: five rendered factors, three injected changes, null updates, and a 4×4 grid over the rotation and threshold of the decision boundary (49 model pairs), with code and protocol. (ii) A linear analysis that separates rotation from offset and predicts what each family sees. (iii) A test of these predictions on CNNs under a protocol fixed in advance: seven confirmed findings, three failed predictions (one of them, on two designs, located by a later registered test) and one rejected hypothesis, all reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique: the ordinary counterfactual of f_{new} or the minimum-norm point with both constraints active. Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing is $u_n - u_o$, the difference of the two models' minimum-norm counterfactuals, the witness is $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference has components in ratio $\tan(\theta/2) < 1$ along $(\hat{w}_o, v)$, i.e. mainly along the new dependence v; the witness has $|u^\star \cdot \hat{w}_o| = \min\{d_n \cos\theta, d_o\}$ and, for $\theta \leq \pi/4$, is dominated by v only if $d_o(\sin\theta + \cos\theta) < d_n$. After projecting out $\hat{w}_o$ all three coincide up to scale, as in Theorem 1 of [1].

On CNNs the last statement holds approximately only with $\hat{w}_o$ taken at the witness (Section 4.2). On linear models the solver's distances agree to 10^{-3} (a stricter pre-declared vector criterion failed at $p = 5$ through the solver's tolerance; the closed forms are KKT-certified). In words, gradient differences see a *rotation* and are blind to an *offset*; witnesses exist only on the side whose margin the offset lowers; for a constant logit offset such as the prior correction [13], they abstain exactly when the shift is below 2κ (Corollary 1 in the proofs). The analysis was formulated after the main experiment and before the grid was run.

Table 1: Injected changes; $k^\star$ is the ground-truth factor (none for the grid cell $(0, 0.5)$).

Change	Adaptation data	$k^\star$	Pairs
Shortcut	colour correlated with label (0.99)	colour	3
Subgroup	labels flipped for size > 0.75	size	3
Threshold	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid, $\alpha = 0$	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$, $t \in \{.5, .53, .56, .6\}$	aspect	8
Grid, $\alpha > 0$	same, $\alpha \in \{.1, .25, .5\}$	colour	24

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0, 1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1). A pair enters only if its change is visible in decisions on validation data (pre-declared thresholds; null pairs need decisive disagreement ≤ 0.02). For the grid we measure the realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions, and analyse at these measured values; all 32 realised the intended rotation, while offsets below about 0.05 were not produced reliably.

Metrics and methods. Per pair we use $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0; $k^\star$ in Table 1). All counterfactual methods share one solver (Adam with a decaying penalty, then SLSQP), the same points and the same budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1] (two variants agree within 0.01); **Grad-diff**, the gradient difference [1]; **Imp-diff**, a permutation-importance difference [4]; **DIG**, a line search after [5]; **DeepXplore**, our search without the proximity term, after [6]; and **CF-new**, the ordinary counterfactual of f_{new} alone.

Protocol. Every test split served at most one confirmatory test, after all choices were fixed. Table 3 lists every test; the grid's manipulation check, exclusion rule and statistics were fixed after a pilot and before its test data; one analysis-code error was later corrected to the registered rule. Bootstraps are paired and stratified by pair; the rest is descriptive. A digit tier and a label-shift update failed pre-declared qualification and were stopped.

4 Results

Findings are grouped by the kind of change; Table 3 lists every test.

Table 2: Recovery@r (found-rate), locked test split, mean over three pairs (95% CIs within ±0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	DIG	DeepXplore	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

Table 3: Tests (95% CIs; †90%). Formed before any result: H2a, P-tie; after the first: C1–C3; before the grid: H5a–H5d; later (new data): H6–H8. Holm: {C1, C3}, {H5a–H5d}, {H7c, H8}. Proj.: $\hat{w}_o$ projected out.

Claim	Data	Estimate	p_{Holm}	Status	ID
Δ-CE > Grad-diff	Subgroup	-0.09 [-0.13, -0.05]		rejected	H2a
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$		not conf.	P-tie
Δ-CE > Grad-diff	Threshold	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$	confirmed	C1
No witness on nulls	Null; Adult	0/2 400; 0/1 500		confirmed	C2, H6
Proj. Δ-CE ≈ Grad-diff	Shortcut	-0.30 [-0.33, -0.28]†		not conf.	C3
Grad-diff < 0.2, offsets	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$	confirmed	H5a
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$	confirmed	H5b
Grad-diff offset-free	Grid	-0.098 [-0.143, -0.053]†		not conf.	H5c
Proj. Δ-CE ≈ Grad-diff	Grid	-0.35 [-0.38, -0.33]†		not conf.	H5d
CF-diff flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$	confirmed	H7c
Witness normal narrows gap $\geq \frac{1}{2}$	Grid	$-0.19 \to +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$	confirmed	H8

4.1 Offsets

Gradient differences are blind to an offset, and witnesses lie on the side the offset lowers, as Proposition 2(a) predicts. On Threshold, Δ-CE beats Grad-diff (0.41 vs. 0.10 on the new test set; Table 2 is the locked split); on the pure offsets of the grid, Grad-diff Recovery is 0.081 (Table 3). The content is the failure of Grad-diff: on an offset the changed factor is the task factor, so even CF-new recovers it (0.90–0.98). Witnesses lie only on the side the offset lowers (on Threshold: every class-1 point, no class-0 point). For an exact offset b on one logit, witnesses appear only once $\lvert b\rvert$ reaches 2κ, then, empirically, on every query of the lowered class only, while CF-diff flags 0.961 of queries already at $\lvert b\rvert \in \{0.5, 0.9\}$, where 3–8% of decisions flip.

4.2 Rotations

Gradient differences see a rotation, increasingly with its angle; after projection, witnesses agree with them only when the old normal is taken at the witness. Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ (Fig. 1), while CF-new recovers colour on ≤ 0.03 of points. Proposition 2(b) predicts that the two coincide up to scale after projecting out $\hat{w}_o$; with $\hat{w}_o$ taken at the query they do not (Shortcut 0.64 vs. 0.95; grid -0.35). A post-hoc explanation, tested on new data (four grid pairs, points with a witness): with $\hat{w}_o$ taken at the witness,

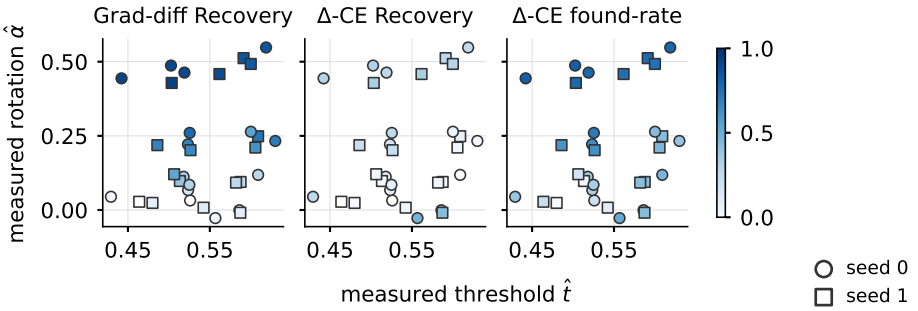


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} (test split), on one colour scale from 0 to 1. Hollow: no $k^\star$; marker shape: seed of f_{old}.

the projected Recovery gap Δ-CE − Grad-diff moves from −0.190 to +0.029. This locates the failure (where the normal is taken), not its cause (why the normal turns). Descriptive: on Shortcut it moves from −0.166 to −0.024; the normal turns by a median 32° from z_0 to $z^\star$ (27° for the nearest quarter of witnesses), so the registered prediction that this grows with $\|z^\star - z_0\|$ failed; the grid gap of −0.35 splits into −0.18 abstention and −0.17 geometry. The original projection tests stay not confirmed.

4.3 Null updates and abstention

Abstention comes from verifying the margin, which all three disagreement searches do: none returns a point on a null update (Table 2). What Δ-CE adds is the witness closest to the query. DeepXplore (Δ-CE without proximity) finds witnesses on fewer points (0.40–0.63 vs. 0.47–0.84); where both find one, the Δ-CE witness is nearer on every point, as the objective requires (median 0.20–0.23 vs. 0.31–0.33; descriptive), and what this buys is higher Recovery on Subgroup and Threshold. No witness of either type exists on the 8 null pairs, while CF-diff attributes a change to 99.1–99.8% of points; with $\kappa = 0$ witnesses reappear (0.80, 0.89), as Lemma 1 predicts, and retraining noise fades smoothly with κ. The same holds for five null updates of an MLP on the numerical Adult features: no witness on 1 500 test points, CF-diff 97.7–100%. NOT_FOUND is not a solver failure: a search of 10^5 points per query finds no witness on 400 null and 150 impossible-class queries (95% bounds 0.9%, 2.4%), and misses a witness on 3 (4 in a re-run) of 395 small-offset queries (descriptive).

4.4 What failed

Δ-CE does not beat gradient differences where we predicted it would, and gradient differences are not fully independent of the offset. On Subgroup, Δ-CE is below Grad-diff (−0.091, negative in all seeds), rejecting the pre-registered hypothesis; it beats CF-diff (+0.21) and DIG (+0.47). The predicted tie on Shortcut and

Threshold fails in all four comparisons. Adding an offset to a rotation lowers Grad-diff Recovery by 0.098 (90% CI $[-0.143, -0.053]$, not within ± 0.10).

5 Discussion and limitations

The linear analysis predicts most of which family sees which kind of change, with the old normal taken at the witness; gradient differences are not fully independent of the offset. A reading not tested as a procedure: witnesses decide *whether* and *where* decisions changed near a query, gradient differences describe a rotation. **Limitations.** (1) The image data are synthetic; the tabular test covers abstention only. (2) Of label shift, only the exact prior correction [13] was tested; an update by fine-tuning failed qualification. (3) κ is an exact threshold only for a constant offset; otherwise abstention tracks decisive change, not a measured shift (at shifts of 0.03–0.04 it depended on the fine-tuning recipe). (4) Witnesses see only decision changes on the manifold within the budget, with no certificate for deep networks; our search for missed witnesses is a sampling bound. (5) Several tests were formed after earlier results, one after a validation run showed the effect (Table 3). Every change to the protocol was logged and made before the test data it concerns were generated, except the analysis-code correction above; the log, proofs and code are at [repository link].

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